

Topological Coordination for a Dual-Gantry Quad-Arm Ceiling Manipulation

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One topological representation, two layers:
the room the robot sees, and the reach each arm owns, priced in energy.

Background



- Aging population → assistive robots **in-home**
- Small, cluttered, **shared with residents**
- **Ceiling-mounted** → floor stays free
- **2 rails × 2 arms** = 4 arms, **8-DOF** each
- **RGB-D only**, no LIDAR

Problems

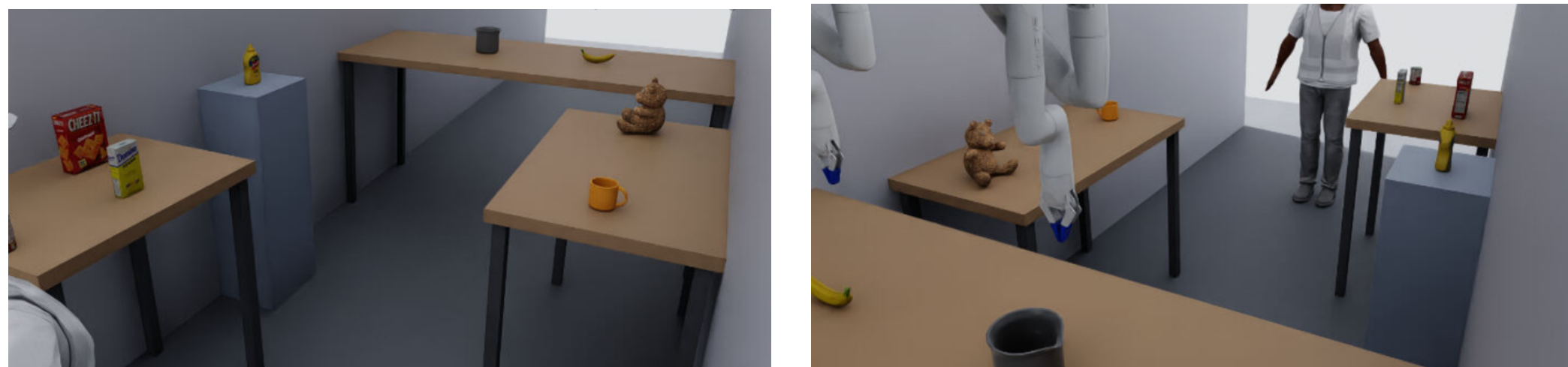
- **P1**: which arm, where the base stops — coupled
- **P2**: fixed-base maps only, reachable / not
- **P3**: occupancy has no topology

Core Idea: One GNG, Two Layers

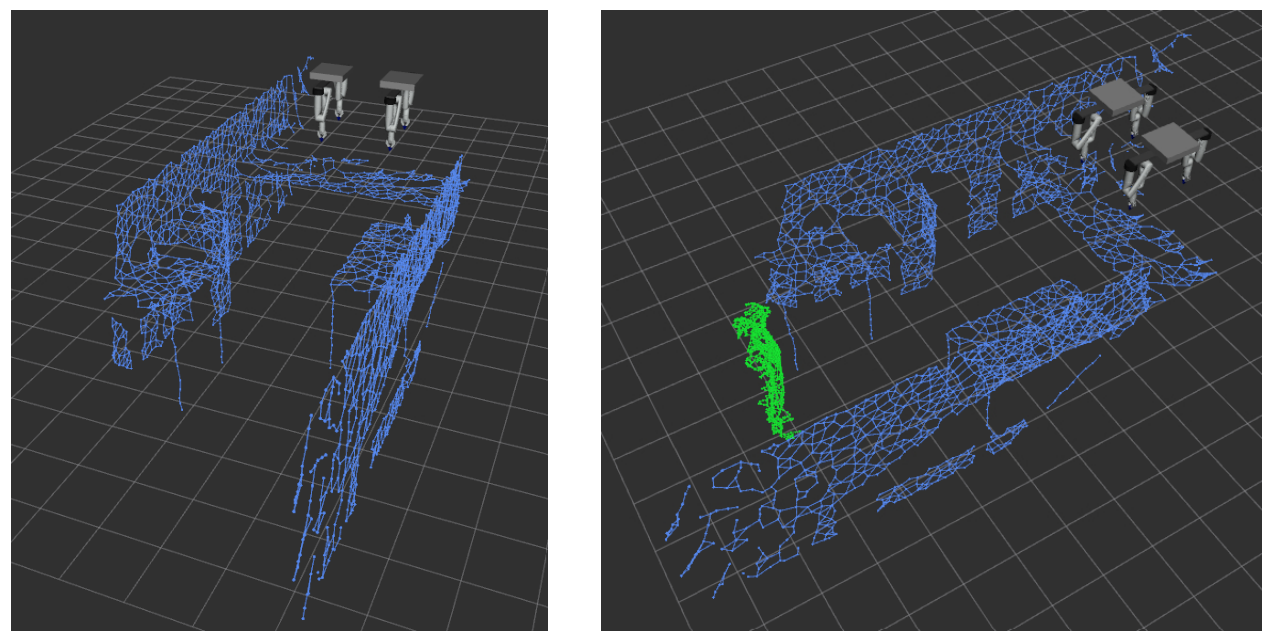
One Growing Neural Gas, fitted twice: what the cameras see (environment) and what each arm can reach (capability).

Layer 1: Environment GNG from RGB-D

- 2 overhead RGB-D cameras, **no LIDAR**
- **Placement**: ceiling-mounted ($z \approx 2.05$ m), diagonally opposite corners, angled down/inward at the table
- `depth_cloud`: depth → world points, no seg. gate
- GNG on the cloud → scene *topology*
- **Static**: frozen once. **Dynamic**: live remainder only
- Union → MoveIt 2 planning scene; tracks by Hungarian match



The two overhead RGB-D views: the entire input (NVIDIA Isaac Sim).

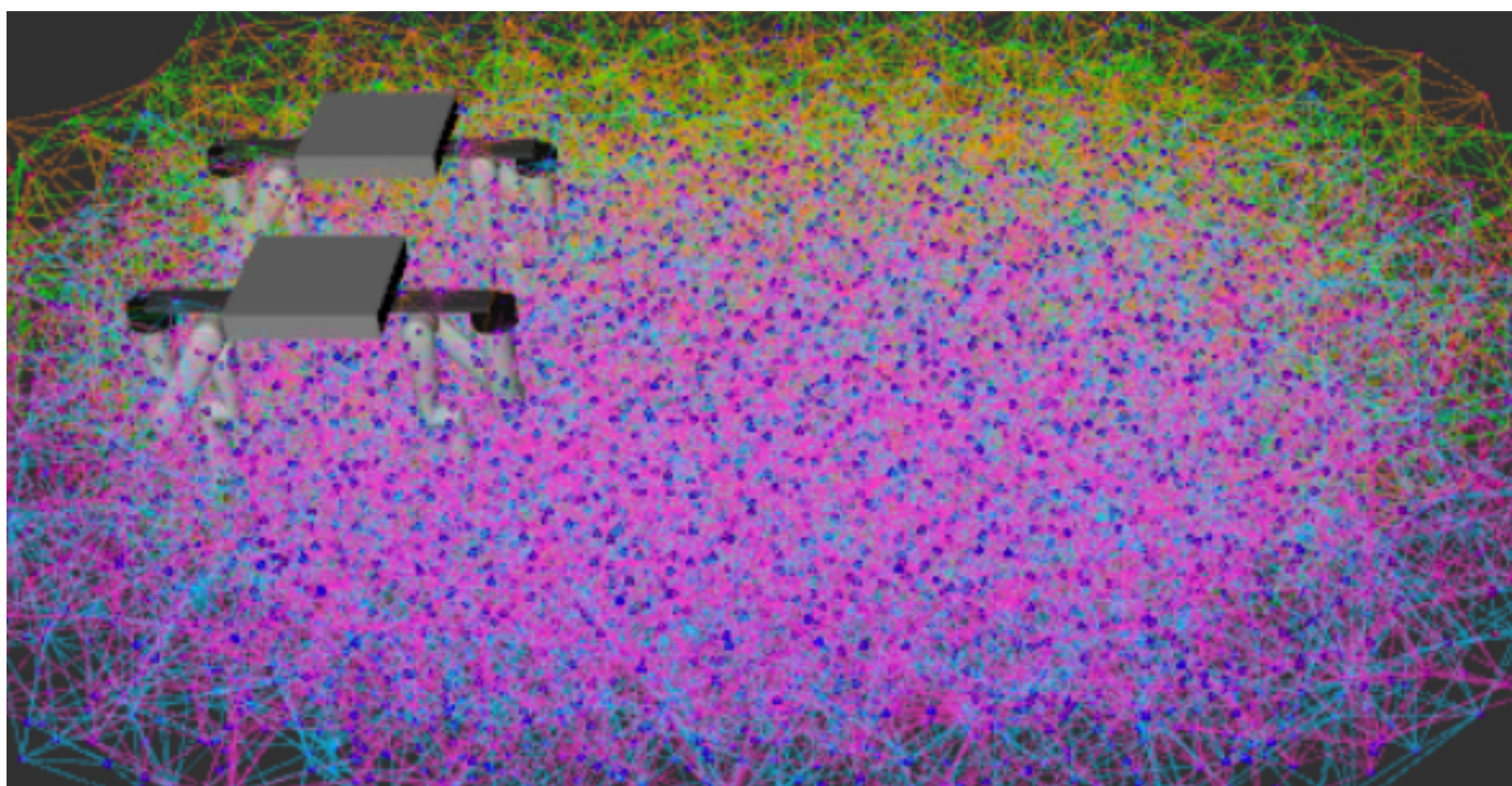


Left: frozen *static* backbone. Right: plus *dynamic* remainder (a person).

Static vs. dynamic budget: **1800**-node static backbone (offline, multi-epoch, captured once) vs. **800**-node live cap (online, single pass per frame) — same GNG, different node budget & training regime per layer.

Layer 2: Base-Aware Capability GNG

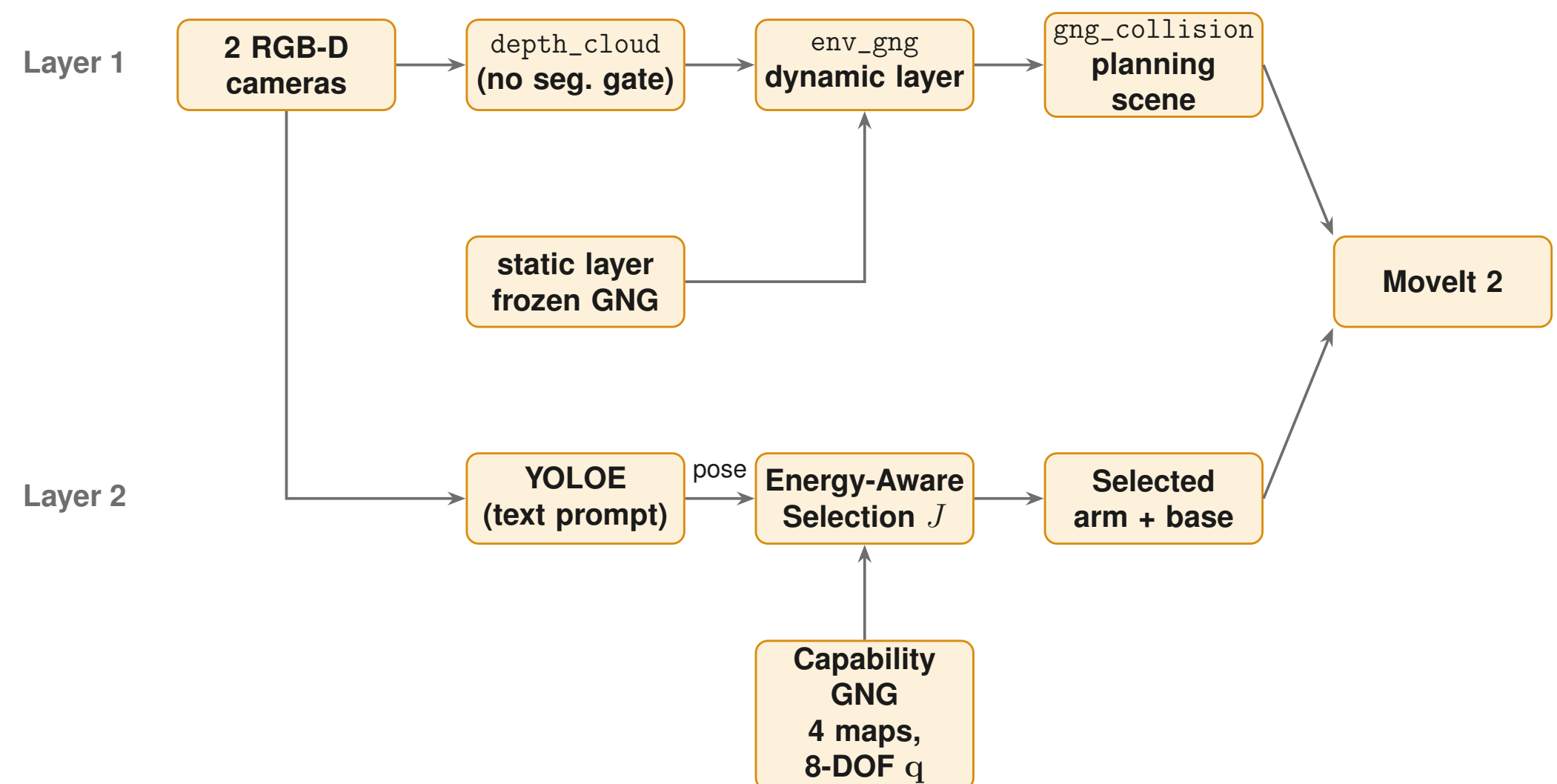
- Sample **full 8-DOF** $\mathbf{q}$, FK (Pinocchio), manipulability $w = \sqrt{\det(JJ^T)}$
- Node stores $[\mathbf{x} \mid \mathbf{q}]$: point + config, **rail included**
- **Boundary seeding** → 4 maps (one per arm)



Capability layer of one arm (RViz), coloured by manipulability.

Per arm: 80,000 FK samples → **2915 nodes** (600 boundary), 14,547 edges, spacing **0.090 m**.

System and Pipeline



Layer 1 (geometry) is decoupled from Layer 2 (identity + decision).

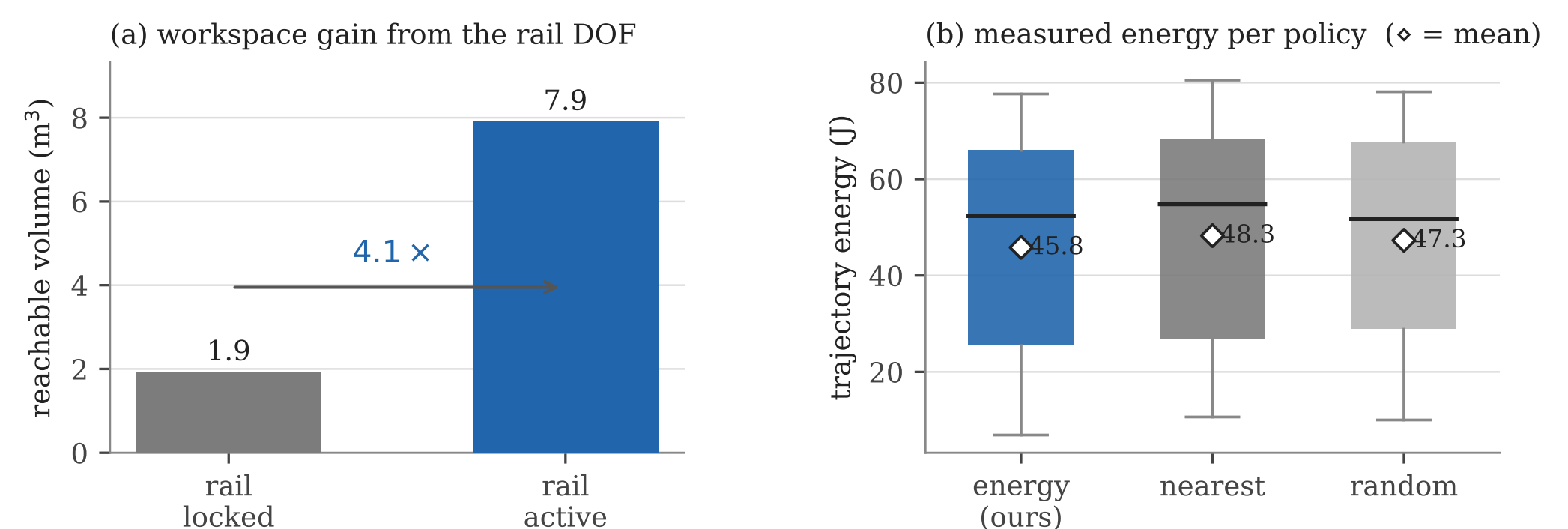
Energy-Aware Selection Cost J

$$J = w_{gl} \frac{\Delta_{lin}}{\rho_{gl}} + w_{gr} \frac{\Delta_{rot}}{\rho_{gr}} + w_{arm} \frac{\Delta_{arm}}{\rho_{arm}} + w_{dist} \frac{e}{\rho_{dist}} - w_{manip} \frac{m}{\rho_{manip}}$$

- Rail/arm travel, tool distance, manipulability — **fit by OLS** on 168 picks ($R^2 = 0.857$): rail dominates
- **GNG** → J : every **capability-layer node** near the target (all 4 arms) scored by J from its own $\mathbf{q}$
- **Selection**: **lowest- J first, plan-only**; first collision-free plan wins

Experiments and Results

- **Feasibility, not just efficiency**: **93.3%** multi-arm success vs. **63–65%** for any single fixed arm
- **Why**: rail active → $4\times$ reachable volume (**1.9**→**7.9 m³**)
- **Boundary seeding**: reach-surface error **0.221 m** → **0.008 m**



(a) Reachable volume. (b) Trajectory energy per policy ($\diamond$ =mean).

Selection policies, 60 target poses, plan-only.

Mode	Success	Rail lin. (m)	Energy (J)
energy (ours)	56/60 (93.3%)	0.97/1.01	45.84/52.34
nearest	56/60 (93.3%)	1.19/1.41	48.28/54.79
random	56/60 (93.3%)	1.17/1.28	47.31/51.72
fixed (best, arm_4)	39/60 (65.0%)	1.14/1.21	49.79/55.78

Fixed-arm energies use that arm's own smaller reachable set.

- **Energy (secondary)**: lowest mean, but modest (**45.84 J** vs. 48.28 J, -5.1%)
- **Trade-off confirmed**: arm_3 (0.64 m rail) chosen over nearest arm_4 (1.15 m rail): rail cost avoided

End-to-End Validation



- **Full pipeline**: **35/35 (100%)** picks, **39.6 s/pick** avg
- **Perception**: **100%** detection, **0.023 m** error
- **Reachability screening**: **100% (8/8)** correct
- **Environment layer**: **1800** nodes static, **800**-node live cap, ~ 1 Hz

Summary and Contributions

- **Feasibility over efficiency**: **93.3%** vs. **63–65%** success
- **Base as a genuine DOF**: $\approx 4\times$ reachable volume
- **Coupling, correctly priced**: J fit on measured energy ($R^2 = 0.857$)
- **One GNG, two layers**: one representation for collision *and* capability

Future work: real hardware; dynamic-scene study.